Algebraic Geometry(GTM52)

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Chapter 1

Varities

Our purpose in this chapter is to give an introduction to algebraic geometry with as little machinery as possible. We work over a fixed algebraically closed field k. We define the main objects of study, which are algebraic varieties in affine or projective space. We introduce some of the most important concepts, such as dimension, regular functions, rational maps, nonsingular varieties, and the degree of a projective variety. And most im - portant, we give lots of specific examples, in the form of exercises at the end of each section. The examples have been selected to illustrate many interesting and important phenomena, beyond those mentioned in the text. The person who studies these examples carefully will not only have a good understanding of the basic concepts of algebraic geometry, but he will also have the background to appreciate some of the more abstract developments of modern algebraic geometry, and he will have a resource against which to check his intuition. We will continually refer back to this library of examples in the rest of the book.

The last section of this chapter is a kind of second introduction to the book. It contains a discussion of the "classification problem," which has motivated much of the development of algebraic geometry. It also contains a discussion of the degree of generality in which one should develop the foundations of algebraic geometry, and as such provides motivation for the theory of schemes.

1.1 Affine Varieties

Let k be a fixed algebraically closed field. We define *affine n-space* over k, denoted $\mathbf{A}^n_k$ or simply $\mathbf{A}^n$, to be the set of all n-tuples of elements of k. An element $P \in \mathbf{A}^n$ will be called a *point*, and if $P = (a_1, \ldots, a_n)$ with $a_i \in k$, then the a_i will be called the *coordinates* of P.

Let $A = k[x_1, \ldots, x_n]$ be the polynomial ring in n variables over k. We will interpret the elements of A as functions from the affine n-space to k, by defining $f(P) = f(a_1, \ldots, a_n)$, where $f \in A$ and $P \in \mathbb{A}^n$. Thus if $f \in A$ is a polynomial, we can talk about the set of zeros of f, namely $Z(f) = \{P \in \mathbb{A}^n | f(P) = 0\}$. More generally, if T is any subset of A, we define the zero - set of T to be the common zeros of all the elements of T, namely

$$Z(T) = \{P \in \mathbb{A}^n | f(P) = 0 \text{ for all } f \in T\}.$$

Clearly if $\mathfrak{a}$ is the ideal of A generated by T, then $Z(T) = Z(\mathfrak{a})$. Furthermore, since A is

a noetherian ring, any ideal $\mathfrak{a}$ has a finite set of generators $f_1, \ldots, f_r$. Thus $Z(T)$ can be expressed as the common zeros of the finite set of polynomials $f_1, \ldots, f_r$.

Definition 1.1.1. A subset Y of $\mathbb{A}^n$ is an algebraic set if there exists a subset $T \subseteq A$ such that $Y = Z(T)$.

Proposition 1.1.2. *The union of two algebraic sets is an algebraic set. The intersection of any family of algebraic sets is an algebraic set. The empty set and the whole space are algebraic sets.*

Proof. If $Y_1 = Z(T_1)$ and $Y_2 = Z(T_2)$, then $Y_1 \cup Y_2 = Z(T_1T_2)$, where T_1T_2 denotes the set of all products of an element of T_1 by an element of T_2. Indeed, if $P \in Y_1 \cup Y_2$, then either $P \in Y_1$ or $P \in Y_2$, so P is a zero of every polynomial in T_1T_2. Conversely, if $P \in Z(T_1T_2)$, and $P \notin Y_1$, say, then there is an $f \in T_1$ such that $f(P) \neq 0$. Now for any $g \in T_2$, $(fg)(P) = 0$ implies that $g(P) = 0$, so that $P \in Y_2$.

If $Y_\alpha = Z(T_\alpha)$ is any family of algebraic sets, then $\bigcap Y_\alpha = Z(\bigcup T_\alpha)$, so $\bigcap Y_\alpha$ is also an algebraic set. Finally, the empty set $\varnothing = Z(1)$, and the whole space $\mathbb{A}^n = Z(0)$. $\square$

Definition 1.1.3. We define the Zariski topology on $\mathbb{A}^n$ by taking the open subsets to be the complements of the algebraic sets. This is a topology, because according to the proposition, the intersection of two open sets is open, and the union of any family of open sets is open. Furthermore, the empty set and the whole space are both open.

Example 1.1.4. Let us consider the Zariski topology on the affine line $\mathbb{A}^1$. Every ideal in $A = k[x]$ is principal, so every algebraic set is the set of zeros of a single polynomial. Since k is algebraically closed, every non - zero polynomial $f(x)$ can be written $f(x) = c(x-a_1)\cdots(x-a_n)$ with $c, a_1, \ldots, a_n \in k$. Then $Z(f) = \{a_1, \ldots, a_n\}$. Thus the algebraic sets in $\mathbb{A}^1$ are just the finite subsets (including the empty set) and the whole space (corresponding to $f = 0$). Thus the open sets are the empty set and the complements of finite subsets. Notice in particular that this topology is not Hausdorff.

Definition 1.1.5. A nonempty subset Y of a topological space X is irreducible if it cannot be expressed as the union $Y = Y_1 \cup Y_2$ of two proper subsets, each one of which is closed in Y. The empty set is not considered to be irreducible.

Example 1.1.6. $\mathbb{A}^1$ is irreducible, because its only proper closed subsets are finite, yet it is infinite (because k is algebraically closed, hence infinite).

Example 1.1.7. Any nonempty open subset of an irreducible space is irreducible and dense.

Example 1.1.8. If Y is an irreducible subset of X, then its closure $\overline{Y}$ in X is also irreducible.

Definition 1.1.9. An affine algebraic variety (or simply affine variety) is an irreducible closed subset of $\mathbb{A}^n$ (with the induced topology). An open subset of an affine variety is a quasi - affine variety.

These affine and quasi - affine varieties are our first objects of study. But before we can go further, in fact before we can even give any interesting examples, we need to explore the

relationship between subsets of $\mathbb{A}^n$ and ideals in A more deeply. So for any subset $Y \subseteq \mathbb{A}^n$, let us define the ideal of Y in A by

$$I(Y) = \{f \in A | f(P) = 0 \text{ for all } P \in Y\}.$$

Now we have a function Z which maps subsets of A to algebraic sets, and a function I which maps subsets of $\mathbb{A}^n$ to ideals. Their properties are summarized in the following proposition.

Proposition 1.1.10. *(a) If $T_1 \subseteq T_2$ are subsets of A, then $Z(T_1) \supseteq Z(T_2)$.*
(b) If $Y_1 \subseteq Y_2$ are subsets of $\mathbb{A}^n$, then $I(Y_1) \supseteq I(Y_2)$.
(c) For any two subsets Y_1, Y_2 of $\mathbb{A}^n$, we have $I(Y_1 \cup Y_2) = I(Y_1) \cap I(Y_2)$.
(d) For any ideal $\mathfrak{a} \subseteq A$, $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$, the radical of $\mathfrak{a}$.
(e) For any subset $Y \subseteq \mathbb{A}^n$, $Z(I(Y)) = \overline{Y}$, the closure of Y.

Proof. (a), (b) and (c) are obvious. (d) is a direct consequence of Hilbert's Nullstellensatz, stated below, since the radical of $\mathfrak{a}$ is defined as

$$\sqrt{\mathfrak{a}} = \{f \in A | f^r \in \mathfrak{a} \text{ for some } r > 0\}.$$

To prove (e), we note that $Y \subseteq Z(I(Y))$, which is a closed set, so clearly $\overline{Y} \subseteq Z(I(Y))$. On the other hand, let W be any closed set containing Y. Then $W = Z(\mathfrak{a})$ for some ideal $\mathfrak{a}$. So $Z(\mathfrak{a}) \supseteq Y$, and by (b), $I(Z(\mathfrak{a})) \subseteq I(Y)$. But certainly $\mathfrak{a} \subseteq IZ(\mathfrak{a})$, so by (a) we have $W = Z(\mathfrak{a}) \supseteq ZI(Y)$. Thus $ZI(Y) = \overline{Y}$. □

Theorem 1.1.11 (Hilbert's Nullstellensatz)**.** *Let k be an algebraically closed field, let $\mathfrak{a}$ be an ideal in $A = k[x_1, \ldots, x_n]$, and let $f \in A$ be a polynomial which vanishes at all points of $Z(\mathfrak{a})$. Then $f^r \in \mathfrak{a}$ for some integer $r > 0$.*

Proof. Lang or Atiyah–Macdonald □

Corollary 1.1.12. *There is a one - to - one inclusion - reversing correspondence between algebraic sets in $\mathbb{A}^n$ and radical ideals (i.e., ideals which are equal to their own radical) in A, given by $Y \mapsto I(Y)$ and $\mathfrak{a} \mapsto Z(\mathfrak{a})$. Furthermore, an algebraic set is irreducible if and only if its ideal is a prime ideal.*

Proof. Only the last part is new. If Y is irreducible, we show that $I(Y)$ is prime. Indeed, if $fg \in I(Y)$, then $Y \subseteq Z(fg) = Z(f) \cup Z(g)$. Thus $Y = (Y \cap Z(f)) \cup (Y \cap Z(g))$, both being closed subsets of Y. Since Y is irreducible, we have either $Y = Y \cap Z(f)$, in which case $Y \subseteq Z(f)$, or $Y \subseteq Z(g)$. Hence either $f \in I(Y)$ or $g \in I(Y)$.

Conversely, let $\mathfrak{p}$ be a prime ideal, and suppose that $Z(\mathfrak{p}) = Y_1 \cup Y_2$. Then $\mathfrak{p} = I(Y_1) \cap I(Y_2)$, so either $\mathfrak{p} = I(Y_1)$ or $\mathfrak{p} = I(Y_2)$. Thus $Z(\mathfrak{p}) = Y_1$ or Y_2, hence it is irreducible. □

Example 1.1.13. $\mathbb{A}^n$ is irreducible, since it corresponds to the zero ideal in A, which is prime.

Example 1.1.14. Let f be an irreducible polynomial in $A = k[x, y]$. Then f generates a prime ideal in A, since A is a unique factorization domain, so the zero set $Y = Z(f)$ is irreducible. We call it the affine curve defined by the equation $f(x, y) = 0$. If f has degree d, we say that Y is a curve of degree d.

Example 1.1.15. More generally, if f is an irreducible polynomial in $A = k[x_1, \ldots, x_n]$, we obtain an affine variety $Y = Z(f)$, which is called a surface if $n = 3$, or a hypersurface if $n > 3$.

Example 1.1.16. A maximal ideal $\mathfrak{m}$ of $A = k[x_1, \ldots, x_n]$ corresponds to a minimal irreducible closed subset of $\mathbb{A}^n$, which must be a point, say $P = (a_1, \ldots, a_n)$. This shows that every maximal ideal of A is of the form $\mathfrak{m} = (x_1 - a_1, \ldots, x_n - a_n)$, for some $a_1, \ldots, a_n \in k$.

Example 1.1.17. If k is not algebraically closed, these results do not hold. For example, if $k = \mathbb{R}$, the curve $x^2 + y^2 + 1 = 0$ in $\mathbb{A}^2_{\mathbb{R}}$ has no points. So (1.2d) is false. See also (Ex. 1.12).

Definition 1.1.18. If $Y \subseteq \mathbb{A}^n$ is an affine algebraic set, we define the affine coordinate ring $A(Y)$ of Y to be $A/I(Y)$.

Remark 1.1.19. If Y is an affine variety, then $A(Y)$ is an integral domain. Furthermore, $A(Y)$ is a finitely generated k-algebra. Conversely, any

Remark 1.1.20. If Y is an affine variety, then $A(Y)$ is an integral domain and finitely - generated k-algebra. Conversely, any finitely generated k-algebra B which is a domain is the affine coordinate ring of some affine variety. Indeed, write B as the quotient of a polynomial ring $A = k[x_1, \ldots, x_n]$ by an ideal $\mathfrak{a}$, and let $Y = Z(\mathfrak{a})$.

Next we will study the topology of our varieties. To do so we introduce an important class of topological spaces which includes all varieties.

Definition 1.1.21. A topological space X is called noetherian if it satisfies the descending chain condition for closed subsets: for any sequence $Y_1 \supseteq Y_2 \supseteq \ldots$ of closed subsets, there is an integer r such that $Y_r = Y_{r+1} = \ldots$.

Example 1.1.22. $\mathbb{A}^n$ is a noetherian topological space. Indeed, if $Y_1 \supseteq Y_2 \supseteq \ldots$ is a descending chain of closed subsets, then $I(Y_1) \subseteq I(Y_2) \subseteq \ldots$ is an ascending chain of ideals in $A = k[x_1, \ldots, x_n]$. Since A is a noetherian ring, this chain of ideals is eventually stationary. But for each i, $Y_i = Z(I(Y_i))$, so the chain Y_i is also stationary.

Proposition 1.1.23. *In a noetherian topological space X, every non - empty closed subset Y can be expressed as a finite union $Y = Y_1 \cup \ldots \cup Y_r$ of irreducible closed subsets Y_i. If we require that $Y_i \nsubseteq Y_j$ for $i \neq j$, then the Y_i are uniquely determined. They are called the irreducible components of Y.*

Proof. First we show the existence of such a representation of Y. Let $\mathfrak{S}$ be the set of non - empty closed subsets of X which cannot be written as a finite union of irreducible closed subsets. If $\mathfrak{S}$ is non - empty, then since X is noetherian, it must contain a minimal element, say Y. Then Y is not irreducible, by construction of $\mathfrak{S}$. Thus we can write $Y = Y' \cup Y''$, where Y' and Y'' are proper closed subsets of Y. By minimality of Y, each of Y' and Y'' can be expressed as a finite union of closed irreducible subsets, hence Y also, which is a contradiction. We conclude that every closed set Y can be written as a union $Y = Y_1 \cup \ldots \cup Y_r$ of irreducible subsets. By throwing away a few if necessary, we may assume $Y_i \nsubseteq Y_j$ for $i \neq j$.

Now suppose $Y = Y_1' \cup \ldots \cup Y_s'$ is another such representation. Then $Y_1' \subseteq Y = Y_1 \cup \ldots \cup Y_r$, so $Y_1' = \bigcup (Y_1' \cap Y_i)$. But Y_1' is irreducible, so $Y_1' \subseteq Y_i$ for some i, say $i = 1$. Similarly, $Y_1 \subseteq Y_j'$ for some j. Then $Y_1' \subseteq Y_j'$, so $j = 1$, and we find that $Y_1 = Y_1'$. Now let $Z = (Y - Y_1)^-$. Then $Z = Y_2 \cup \ldots \cup Y_r$ and also $Z = Y_2' \cup \ldots \cup Y_s'$. So proceeding by induction on r, we obtain the uniqueness of the Y_i. $\square$

Corollary 1.1.24. *Every algebraic set in $\mathbb{A}^n$ can be expressed uniquely as a union of varieties, no one containing another.*

Definition 1.1.25. If X is a topological space, we define the dimension of X (denoted $\dim X$) to be the supremum of all integers n such that there exists a chain $Z_0 \subset Z_1 \subset \ldots \subset Z_n$ of distinct irreducible closed subsets of X. We define the dimension of an affine or quasi - affine variety to be its dimension as a topological space.

Example 1.1.26. The dimension of $\mathbb{A}^1$ is 1. Indeed, the only irreducible closed subsets of $\mathbb{A}^1$ are the whole space and single points.

Definition 1.1.27. In a ring A, the height of a prime ideal $\mathfrak{p}$ is the supremum of all integers n such that there exists a chain $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \ldots \subset \mathfrak{p}_n = \mathfrak{p}$ of distinct prime ideals. We define the dimension (or Krull dimension) of A to be the supremum of the heights of all prime ideals.

Proposition 1.1.28. *If Y is an affine algebraic set, then the dimension of Y is equal to the dimension of its affine coordinate ring $A(Y)$.*

Proof. If Y is an affine algebraic set in $\mathbb{A}^n$, then the closed irreducible subsets of Y correspond to prime ideals of $A = k[x_1, \ldots, x_n]$ containing $I(Y)$. These in turn correspond to prime ideals of $A(Y)$. Hence $\dim Y$ is the length of the longest chain of prime ideals in $A(Y)$, which is its dimension. □

This proposition allows us to apply results from the dimension theory of noetherian rings to algebraic geometry.

Theorem 1.1.29. *Let k be a field, and let B be an integral domain which is a finitely - generated k-algebra. Then:*

1. *the dimension of B is equal to the transcendence degree of the quotient field $K(B)$ of B over k;*
2. *For any prime ideal $\mathfrak{p}$ in B, we have height $\mathfrak{p} + \dim B/\mathfrak{p} = \dim B$.*

Proof. Matsumura [2, Ch. 5, §14] or, in the case k is algebraically closed, Atiyah - Macdonald [1, Ch. 11]. □

Proposition 1.1.30. *The dimension of $\mathbb{A}^n$ is n.*

Proof. According to (1.7) this says that the dimension of the polynomial ring $k[x_1, \ldots, x_n]$ is n, which follows from part (a) of the theorem. □

Proposition 1.1.31. *If Y is a quasi - affine variety, then $\dim Y = \dim \overline{Y}$.*

Proof. If $Z_0 \subset Z_1 \subset \ldots \subset Z_n$ is a sequence of distinct closed irreducible subsets of Y, then $\overline{Z}_0 \subset \overline{Z}_1 \subset \ldots \subset \overline{Z}_n$ is a sequence of distinct closed irreducible subsets of $\overline{Y}$ (1.1.4), so we have $\dim Y \leqslant \dim \overline{Y}$. In particular, $\dim Y$ is finite, so we can choose a maximal such chain $Z_0 \subset \ldots \subset Z_n$, with $n = \dim Y$. In that case Z_0 must be a point P, and the chain $P = \overline{Z}_0 \subset \ldots \subset \overline{Z}_n$ will also be maximal (1.1.3). Now P corresponds to a maximal ideal $\mathfrak{m}$ of the affine coordinate ring $A(\overline{Y})$ of $\overline{Y}$. The $\overline{Z}_i$ correspond to prime ideals contained in $\mathfrak{m}$, so height $\mathfrak{m} = n$. On the other hand, since P is a point in affine space, $A(\overline{Y})/\mathfrak{m} \cong k$ (1.4.4). Hence by (1.8Ab) we find that $n = \dim A(\overline{Y}) = \dim \overline{Y}$. Thus $\dim Y = \dim \overline{Y}$. □

Theorem 1.1.32 (Krull's Hauptidealsatz)**.** *Let A be a noetherian ring, and let $f \in A$ be an element which is neither a zero - divisor nor a unit. Then every minimal prime ideal $\mathfrak{p}$ containing f has height* 1.

Proof. Atiyah - Macdonald [1, p. 122]. □

Proposition 1.1.33. *A noetherian integral domain A is a unique factorization domain if and only if every prime ideal of height* 1 *is principal.*

Proof. Matsumura [2, p. 141], or Bourbaki [1, Ch. 7, §3]. □

Proposition 1.1.34. *A variety Y in $\mathbb{A}^n$ has dimension $n-1$ if and only if it is the zero set $Z(f)$ of a single non - constant irreducible polynomial in $A = k[x_1, \ldots, x_n]$.*

Proof. If f is an irreducible polynomial, we have already seen that $Z(f)$ is a variety. Its ideal is the prime ideal $\mathfrak{p} = (f)$. By (1.11A), $\mathfrak{p}$ has height 1, so by (1.8A), $Z(f)$ has dimension $n-1$. Conversely, a variety of dimension $n-1$ corresponds to a prime ideal of height 1. Now the polynomial ring A is a unique factorization domain, so by (1.12A), $\mathfrak{p}$ is principal, necessarily generated by an irreducible polynomial f. Hence $Y = Z(f)$. □

Remark 1.1.35. A prime ideal of height 2 in a polynomial ring cannot necessarily be generated by two elements (Ex. 1.11).

Exercises

1.1 (a) Let Y be the plane curve $y = x^2$ (i.e., Y is the zero set of the polynomial $f = y - x^2$). Show that $A(Y)$ is isomorphic to a polynomial ring in one variable over k.

(b) Let Z be the plane curve $xy = 1$. Show that $A(Z)$ is not isomorphic to a polynomial ring in one variable over k.

*(c) Let f be any irreducible quadratic polynomial in $k[x, y]$, and let W be the conic defined by f. Show that $A(W)$ is isomorphic to $A(Y)$ or $A(Z)$. Which one is it when?

1.2 The Twisted Cubic Curve. Let $Y \subseteq \mathbb{A}^3$ be the set $Y = \{(t, t^2, t^3) | t \in k\}$. Show that Y is an affine variety of dimension 1. Find generators for the ideal $I(Y)$. Show that $A(Y)$ is isomorphic to a polynomial ring in one variable over k. We say that Y is given by the parametric representation $x = t, y = t^2, z = t^3$.

1.3 Let Y be the algebraic set in $\mathbb{A}^3$ defined by the two polynomials $x^2 - yz$ and $xz - x$. Show that Y is a union of three irreducible components. Describe them and find their prime ideals.

1.4 If we identify $\mathbb{A}^2$ with $\mathbb{A}^1 \times \mathbb{A}^1$ in the natural way, show that the Zariski topology on $\mathbb{A}^2$ is not the product topology of the Zariski topologies on the two copies of $\mathbb{A}^1$.

1.5 Show that a k-algebra B is isomorphic to the affine coordinate ring of some algebraic set in $\mathbb{A}^n$, for some n, if and only if B is a finitely generated k-algebra with no nilpotent elements.

1.6 Any nonempty open subset of an irreducible topological space is dense and irreducible. If Y is a subset of a topological space X, which is irreducible in its induced topology, then the closure $\overline{Y}$ is also irreducible.

1.7 (a) Show that the following conditions are equivalent for a topological space X:

(i) X is noetherian;

(ii) every nonempty family of closed subsets has a minimal element;

(iii) X satisfies the ascending chain condition for open subsets;

(iv) every nonempty family of open subsets has a maximal element.

(b) A noetherian topological space is quasi - compact, i.e., every open cover has a finite subcover.

(c) Any subset of a noetherian topological space is noetherian in its induced topology.

(d) A noetherian space which is also Hausdorff must be a finite set with the discrete topology.

1.8 Let Y be an affine variety of dimension r in $\mathbb{A}^n$. Let H be a hypersurface in $\mathbb{A}^n$, and assume that $Y \nsubseteq H$. Then every irreducible component of $Y \cap H$ has dimension $r-1$. (See (7.1) for a generalization.)

1.9 Let $\mathfrak{a} \subseteq A = k[x_1, \ldots, x_n]$ be an ideal which can be generated by r elements. Then every irreducible component of $Z(\mathfrak{a})$ has dimension $\geq n-r$.

1.10 (a) If Y is any subset of a topological space X, then $\dim Y \leq \dim X$.

(b) If X is a topological space which is covered by a family of open subsets $\{U_i\}$, then $\dim X = \sup \dim U_i$.

(c) Give an example of a topological space X and a dense open subset U with $\dim U < \dim X$.

(d) If Y is a closed subset of an irreducible finite - dimensional topological space X, and if $\dim Y = \dim X$, then $Y = X$.

(e) Give an example of a noetherian topological space of infinite dimension.

*1.11 Let $Y \subseteq \mathbb{A}^3$ be the curve given parametrically by $x = t^3$, $y = t^4$, $z = t^5$. Show that $I(Y)$ is a prime ideal of height 2 in $k[x, y, z]$ which cannot be generated by 2 elements. We say Y is not a local complete intersection—cf. (Ex. 2.17).

1.12 Give an example of an irreducible polynomial $f \in \mathbb{R}[x, y]$, whose zero set $Z(f)$ in $\mathbb{A}^2_{\mathbb{R}}$ is not irreducible (cf. 1.4.2).

1.2 Projective Varieties

To define projective varieties, we proceed in a manner analogous to the definition of affine varieties, except that we work in projective space.

Let k be our fixed algebraically closed field. We defined projective n-space over k, denoted $\mathbb{P}^n_k$, or simply $\mathbb{P}^n$, to be the set of equivalence classes of $(n+1)$-tuples $(a_0, \ldots, a_n)$ of elements

of k, not all zero, under the equivalence relation given by $(a_0, \ldots, a_n) \sim (\lambda a_0, \ldots, \lambda a_n)$ for all $\lambda \in k$, $\lambda \neq 0$. Another way of saying this is that $\mathbb{P}^n$ as a set is the quotient of the set

$\mathbb{A}^{n+1} - \{(0, \ldots, 0)\}$ under the equivalence relation which identifies points lying on the same line through the origin.

An element of $\mathbb{P}^n$ is called a point. If P is a point, then any $(n+1)$ - tuple $(a_0, \ldots, a_n)$ in the equivalence class P is called a set of homogeneous coordinates for P.

Let S be the polynomial ring $k[x_0, \ldots, x_n]$. We want to regard S as a graded ring, so we recall briefly the notion of a graded ring.

A graded ring is a ring S, together with a decomposition $S = \bigoplus_{d \geq 0} S_d$ of S into a direct sum of abelian groups S_d, such that for any $d, e \geq 0$, $S_d \cdot S_e \subseteq S_{d+e}$. An element of S_d is called a homogeneous element of degree d. Thus any element of S can be written uniquely as a (finite) sum of homogeneous elements. An ideal $\mathfrak{a} \subseteq S$ is a homogeneous ideal if $\mathfrak{a} = \bigoplus_{d \geq 0} (\mathfrak{a} \cap S_d)$. We will need a few basic facts about homogeneous ideals (see, for example, Matsumura [2, §10] or Zariski - Samuel [1, vol. 2, Ch. VII, §2]). An ideal is homogeneous if and only if it can be generated by homo - geneous elements. The sum, product, intersection, and radical of homo - geneous ideals are homogeneous. To test whether a homogeneous ideal is prime, it is sufficient to show for any two homogeneous elements f, g, that $fg \in \mathfrak{a}$ implies $f \in \mathfrak{a}$ or $g \in \mathfrak{a}$.

We make the polynomial ring $S = k[x_0, \ldots, x_n]$ into a graded ring by taking S_d to be the set of all linear combinations of monomials of total weight d in $x_0, \ldots, x_n$. If $f \in S$ is a polynomial, we cannot use it to define a function on $\mathbb{P}^n$, because of the nonuniqueness of the homogeneous co - ordinates. However, if f is a homogeneous polynomial of degree d, then $f(\lambda a_0, \ldots, \lambda a_n) = \lambda^d f(a_0, \ldots, a_n)$, so that the property of f being zero or not depends only on the equivalence class of $(a_0, \ldots, a_n)$. Thus f gives a function from $\mathbb{P}^n$ to $\{0, 1\}$ by $f(P) = 0$ if $f(a_0, \ldots, a_n) = 0$, and $f(P) = 1$ if $f(a_0, \ldots, a_n) \neq 0$.

Thus we can talk about the zeros of a homogeneous polynomial, namely $Z(f) = \{P \in \mathbb{P}^n | f(P) = 0\}$. If T is any set of homogeneous elements of S, we define the zero set of T to be

$$Z(T) = \{P \in \mathbb{P}^n | f(P) = 0 \text{ for all } f \in T\}.$$

If $\mathfrak{a}$ is a homogeneous ideal of S, we define $Z(\mathfrak{a}) = Z(T)$, where T is the set of all homogeneous elements in $\mathfrak{a}$. Since S is a noetherian ring, any set of homogeneous elements T has a finite subset $f_1, \ldots, f_r$ such that $Z(T) = Z(f_1, \ldots, f_r)$.

Definition 1.2.1. A subset Y of $\mathbb{P}^n$ is an algebraic set if there exists a set T of ho - mogeneous elements of S such that $Y = Z(T)$.

Proposition 1.2.2. *The union of two algebraic sets is an algebraic set. The intersection of any family of algebraic sets is an algebraic set. The empty set and the whole space are algebraic sets.*

Proof. Left to reader (it is similar to the proof of (1.1) above). □

Definition 1.2.3. We define the Zariski topology on $\mathbb{P}^n$ by taking the open sets to be the complements of algebraic sets.

Once we have a topological space, the notions of irreducible subset and the dimension of a subset, which were defined in §1, will apply.

Definition 1.2.4. A projective algebraic variety (or simply projective variety) is an irreducible algebraic set in $\mathbb{P}^n$, with the induced topology. An open subset of a projective variety is a quasi - projective variety. The dimension of a projective or quasi - projective variety is its dimension as a topological space.

If Y is any subset of $\mathbb{P}^n$, we define the homogeneous ideal of Y in S, denoted $I(Y)$, to be the ideal generated by $\{f \in S | f$ is homogeneous and $f(P) = 0$ for all $P \in Y\}$. If Y is an algebraic set, we define the homogeneous coordinate ring of Y to be $S(Y) = S/I(Y)$. We refer to (Ex. 2.1 - 2.7) below for various properties of algebraic sets in projective space and their homogeneous ideals.

Our next objective is to show that projective n-space has an open covering by affine n-spaces, and hence that every projective (respectively, quasi - projective) variety has an open covering by affine (respectively, quasi - affine) varieties. First we introduce some notation.

If $f \in S$ is a linear homogeneous polynomial, then the zero set of f is called a hyperplane. In particular we denote the zero set of x_i by H_i, for $i = 0, \ldots, n$. Let U_i be the open set $\mathbb{P}^n - H_i$. Then $\mathbb{P}^n$ is covered by the open sets U_i, because if $P = (a_0, \ldots, a_n)$ is a point, then at least one $a_i \neq 0$, hence $P \in U_i$. We define a mapping $\varphi_i : U_i \to \mathbb{A}^n$ as follows: if $P = (a_0, \ldots, a_n) \in U_i$, then $\varphi_i(P) = Q$, where Q is the point with affine coordinates

$$\left(\frac{a_0}{a_i}, \ldots, \frac{a_n}{a_i}\right),$$

with a_i/a_i omitted. Note that φ_i is well - defined since the ratios a_j/a_i are independent of the choice of homogeneous coordinates.

Proposition 1.2.5. *The map φ_i is a homeomorphism of U_i with its induced topology to $\mathbb{A}^n$ with its Zariski topology.*

Proof. φ_i is clearly bijective, so it will be sufficient to show that the closed sets of U_i are identified with the closed sets of $\mathbb{A}^n$ by φ_i. We may assume $i = 0$, and we write simply U for U_0 and $\varphi : U \to \mathbb{A}^n$ for φ_0.

Let $A = k[y_1, \ldots, y_n]$. We define a map α from the set S^h of homogeneous elements of S to A, and a map β from $\mathbb{A}^n$ to S^h. Given $f \in S^h$, we set $\alpha(f) = f(1, y_1, \ldots, y_n)$. On the other hand, given $g \in A$ of degree e, then $x_0^e g(x_1/x_0, \ldots, x_n/x_0)$ is a homogeneous polynomial of degree e in the x_i, which we call $\beta(g)$.

Now let $Y \subseteq U$ be a closed subset. Let $\overline{Y}$ be its closure in $\mathbb{P}^n$. This is an algebraic set, so $\overline{Y} = Z(T)$ for some subset $T \subseteq S^h$. Let $T' = \alpha(T)$. Then straightforward checking shows that $\varphi(Y) = Z(T')$. Conversely, let W be a closed subset of $\mathbb{A}^n$. Then $W = Z(T')$ for some subset T' of A, and one checks easily that $\varphi^{-1}(W) = Z(\beta(T')) \cap U$. Thus φ and φ^{-1} are both closed maps, so φ is a homeomorphism. $\square$

Corollary 1.2.6. *If Y is a projective (respectively, quasi - projective) variety, then Y is covered by the open sets $Y \cap U_i$, $i = 0, \ldots, n$, which are homeomorphic to affine (respectively, quasi - affine) varieties via the mapping φ_i defined above.*

EXERCISES

2.1. Prove that the following conditions are equivalent for a topological space X:

(i) X is noetherian;

(ii) Every nonempty family of closed subsets has a minimal element;

(iii) X satisfies the descending chain condition for closed subsets.

2.2. Let X be a topological space, $Y \subset X$ an irreducible subset. Show that the closure $\overline{Y}$ in X is irreducible.

2.3. Let X be a noetherian topological space, $E \subset X$. Show that E can be written uniquely as a finite union $E = E_1 \cup \cdots \cup E_n$, where the E_i are irreducible and no E_i is contained in E_j for $i \neq j$.

2.4. Let A be a ring, $X = \operatorname{Spec} A$. Show that:

(a) The irreducible components of X correspond to the minimal prime ideals of A;

(b) X is irreducible if and only if the nilradical of A is prime.

2.5. Let X be an affine variety. Show that:

(a) The irreducible components of X correspond to the minimal prime ideals of $A(X)$;

(b) $\dim X = \operatorname{tr.deg}_k K(X)$, where $K(X)$ is the function field of X.

2.6. Show that $A(Y_i)$ can be identified with the subring of elements of degree 0 of the localized ring $S(Y)_{x_i}$. Then show that $S(Y)_{x_i} \cong A(Y_i)[x_i, x_i^{-1}]$. Now use (1.7), (1.8A), and (Ex 1.10), and look at transcendence degrees. Conclude also that $\dim Y = \dim Y_i$ whenever Y_i is nonempty.

2.7. (a) $\dim \mathbb{P}^n = n$.

(b) If $Y \subseteq \mathbb{P}^n$ is a quasi-projective variety, then $\dim Y = \dim \overline{Y}$. [*Hint:* Use (Ex. 2.6) to reduce to (1.10).]

2.8. A projective variety $Y \subseteq \mathbb{P}^n$ has dimension $n-1$ if and only if it is the zero set of a single irreducible homogeneous polynomial f of positive degree. Y is called a hypersurface in $\mathbb{P}^n$.

2.9. **Projective Closure of an Affine Variety.** If $Y \subseteq \mathbb{A}^n$ is an affine variety, we identify $\mathbb{A}^n$ with an open set $U_0 \subseteq \mathbb{P}^n$ by the homeomorphism φ_0. Then we can speak of $\overline{Y}$, the closure of Y in $\mathbb{P}^n$, which is called the projective closure of Y.

(a) Show that $I(\overline{Y})$ is the ideal generated by $\beta(I(Y))$, using the notation of the proof of (2.2).

(b) Let $Y \subseteq \mathbb{A}^3$ be the twisted cubic of (Ex. 1.2). Its projective closure $\overline{Y} \subseteq \mathbb{P}^3$ is called the twisted cubic curve in $\mathbb{P}^3$. Find generators for $I(Y)$ and $I(\overline{Y})$, and use this example to show that if $f_1, \ldots, f_r$ generate $I(Y)$, then $\beta(f_1), \ldots, \beta(f_r)$ do not necessarily generate $I(\overline{Y})$.

2.10. **The Cone Over a Projective Variety (Fig. 1).** Let $Y \subseteq \mathbb{P}^n$ be a nonempty algebraic set, and let $\theta : \mathbb{A}^{n+1} - \{(0, \ldots, 0)\} \to \mathbb{P}^n$ be the map which sends the point with

affine coordinates $(a_0, \ldots, a_n)$ to the point with homogeneous coordinates $(a_0, \ldots, a_n)$. We define the affine cone over Y to be

$$C(Y) = \theta^{-1}(Y) \cup \{(0, \ldots, 0)\}.$$

(a) Show that $C(Y)$ is an algebraic set in $\mathbb{A}^{n+1}$, whose ideal is equal to $I(Y)$, considered as an ordinary ideal in $k[x_0, \ldots, x_n]$.

(b) $C(Y)$ is irreducible if and only if Y is.

(c) $\dim C(Y) = \dim Y + 1$.

Sometimes we consider the projective closure $\overline{C(Y)}$ of $C(Y)$ in $\mathbb{P}^{n+1}$. This is called the projective cone over Y.

2.11. **Linear Varieties in $\mathbb{P}^n$.** A hypersurface defined by a linear polynomial is called a hyperplane.

(a) Show that the following two conditions are equivalent for a variety Y in $\mathbb{P}^n$:

(i) $I(Y)$ can be generated by linear polynomials.

(ii) Y can be written as an intersection of hyperplanes.

In this case we say that Y is a linear variety in $\mathbb{P}^n$.

(b) If Y is a linear variety of dimension r in $\mathbb{P}^n$, show that $I(Y)$ is minimally - generated by $n - r$ linear polynomials.

(c) Let Y, Z be linear varieties in $\mathbb{P}^n$, with $\dim Y = r, \dim Z = s$. If $r + s - n \geq 0$, then $Y \cap Z \neq \varnothing$. Furthermore, if $Y \cap Z \neq \varnothing$, then $Y \cap Z$ is a linear variety of dimension $\geq r + s - n$. (Think of $\mathbb{A}^{n+1}$ as a vector space over k, and work with its subspaces.)

2.12. **The d-Uple Embedding.** For given $n, d > 0$, let $M_0, M_1, \ldots, M_N$ be all the monomials of degree d in the $n + 1$ variables $x_0, \ldots, x_n$, where $N = \binom{n+d}{n} - 1$. We define a mapping $\rho_d : \mathbb{P}^n \to \mathbb{P}^N$ by sending the point $P = (a_0, \ldots, a_n)$ to the point $\rho_d(P) = (M_0(a), \ldots, M_N(a))$ obtained by substituting the a_i in the monomials M_j. This is called the d-uple embedding of $\mathbb{P}^n$ in $\mathbb{P}^N$. For example, if $n = 1, d = 2$, then $N = 2$, and the image Y of the 2 - uple embedding of $\mathbb{P}^1$ in $\mathbb{P}^2$ is a conic.

(a) Let $\theta : k[y_0, \ldots, y_N] \to k[x_0, \ldots, x_n]$ be the homomorphism defined by sending y_i to M_i, and let $\mathfrak{a}$ be the kernel of θ. Then $\mathfrak{a}$ is a homogeneous prime ideal, and so $Z(\mathfrak{a})$ is a projective variety in $\mathbb{P}^N$.

(b) Show that the image of ρ_d is exactly $Z(\mathfrak{a})$. (One inclusion is easy. The other will require some calculation.)

(c) Now show that ρ_d is a homeomorphism of $\mathbb{P}^n$ onto the projective variety $Z(\mathfrak{a})$.

(d) Show that the twisted cubic curve in $\mathbb{P}^3$ (Ex. 2.9) is equal to the 3 - uple embedding of $\mathbb{P}^1$ in $\mathbb{P}^3$, for suitable choice of coordinates.

2.13. Let Y be the image of the 2 - uple embedding of $\mathbb{P}^2$ in $\mathbb{P}^5$. This is the Veronese surface. If $Z \subseteq Y$ is a closed curve (a curve is a variety of dimension 1), show that there exists a hypersurface $V \subseteq \mathbb{P}^5$ such that $V \cap Y = Z$.

2.14. **The Segre Embedding.** Let $\psi : \mathbb{P}^r \times \mathbb{P}^s \to \mathbb{P}^N$ be the map defined by sending the ordered pair $(a_0, \ldots, a_r) \times (b_0, \ldots, b_s)$ to $(\ldots, a_i b_j, \ldots)$ in lexicographic order, where $N = rs + r + s$. Note that ψ is well - defined and injective. It is called the Segre embedding. Show that the image of ψ is a subvariety of $\mathbb{P}^N$. [Hint: Let the homogeneous coordinates of $\mathbb{P}^N$ be $\{z_{ij} \mid i = 0, \ldots, r, j = 0, \ldots, s\}$, and let $\mathfrak{a}$ be the kernel of the homomorphism $\psi^* : k[z_{ij}] \to k[x_0, \ldots, x_r, y_0, \ldots, y_s]$ which sends z_{ij} to $x_i y_j$. Then show that Im $\psi = Z(\mathfrak{a})$.]

2.15. **The Quadric Surface in $\mathbb{P}^3$ (Fig. 2).** Consider the surface Q (a surface is a variety of dimension 2) in $\mathbb{P}^3$ defined by the equation $xy - zw = 0$.

(a) Show that Q is equal to the Segre embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ in $\mathbb{P}^3$, for suitable choice of coordinates.

(b) Show that Q contains two families of lines (a line is a linear variety of dimension 1) $\{L_t\}, \{M_t\}$, each parametrized by $t \in \mathbb{P}^1$, with the properties that if $L_t \neq L_u$, then $L_t \cap L_u = \varnothing$; if $M_t \neq M_u$, $M_t \cap M_u = \varnothing$, and for all t, u, $L_t \cap M_u =$ one point.

(c) Show that Q contains other curves besides these lines, and deduce that the Zariski topology on Q is not homeomorphic via ψ to the product topology on $\mathbb{P}^1 \times \mathbb{P}^1$ (where each $\mathbb{P}^1$ has its Zariski topology).

2.16. (a) The intersection of two varieties need not be a variety. For example, let Q_1 and Q_2 be the quadric surfaces in $\mathbb{P}^3$ given by the equations $x^2 - yw = 0$ and $xy - zw = 0$, respectively. Show that $Q_1 \cap Q_2$ is the union of a twisted cubic curve and a line.

(b) Even if the intersection of two varieties is a variety, the ideal of the intersection may not be the sum of the ideals. For example, let C be the conic in $\mathbb{P}^2$ given by the equation $x^2 - yz = 0$. Let L be the line given by $y = 0$. Show that $C \cap L$ consists of one point P, but that $I(C) + I(L) \neq I(P)$.

2.17. **Complete intersections.** A variety Y of dimension r in $\mathbb{P}^n$ is a (strict) complete intersection if $I(Y)$ can be generated by $n - r$ elements. Y is a set - theoretic complete intersection if Y can be written as the intersection of $n - r$ hypersurfaces.

(a) Let Y be a variety in $\mathbb{P}^n$, let $Y = Z(\mathfrak{a})$, and suppose that $\mathfrak{a}$ can be generated by q elements. Then show that $\dim Y \geq n - q$.

(b) Show that a strict complete intersection is a set - theoretic complete intersection.

* The converse of (b) is false. For example let Y be the twisted cubic curve in $\mathbb{P}^3$ (Ex. 2.9). Show that $I(Y)$ cannot be generated by two elements. On the other hand, find hypersurfaces H_1, H_2 of degrees 2,3 respectively, such that $Y = H_1 \cap H_2$.

** It is an unsolved problem whether every closed irreducible curve in $\mathbb{P}^3$ is a set - theoretic intersection of two surfaces. See Hartshorne [1] and Hartshorne [5, III, §5] for commentary.

1.3 Morphisms

So far we have defined affine and projective varieties, but we have not discussed what mappings are allowed between them. We have not even said when two are isomorphic. In this section

we will discuss the regular functions on a variety, and then define a morphism of varieties. Thus we will have a good category in which to work.

Let Y be a quasi-affine variety in $\mathbb{A}^n$. We will consider functions f from Y to k.

Definition 1.3.1. A function $f : Y \to k$ is regular at a point $P \in Y$ if there is an open neighborhood U with $P \in U \subseteq Y$, and polynomials $g, h \in A = k[x_1, \ldots, x_n]$, such that h is nowhere zero on U, and $f = g/h$ on U. (Here of course we interpret the polynomials as functions on $\mathbb{A}^n$, hence on Y.) We say that f is regular on Y if it is regular at every point of Y.

Lemma 1.3.2. *A regular function is continuous, when k is identified with $\mathbb{A}^1_k$ in its Zariski topology.*

Proof. It is enough to show that f^{-1} of a closed set is closed. A closed set of $\mathbb{A}^1_k$ is a finite set of points, so it is sufficient to show that $f^{-1}(a) = \{P \in Y | f(P) = a\}$ is closed for any $a \in k$. This can be checked locally: a subset Z of a topological space Y is closed if and only if Y can be covered by open subsets U such that $Z \cap U$ is closed in U for each U. So let U be an open set on which f can be represented as g/h, with $g, h \in A$, and h nowhere 0 on U. Then $f^{-1}(a) \cap U = \{P \in U | g(P)/h(P) = a\}$. But $g(P)/h(P) = a$ if and only if $(g - ah)(P) = 0$. So $f^{-1}(a) \cap U = Z(g - ah) \cap U$ which is closed. Hence $f^{-1}(a)$ is closed in Y. $\square$

Now let us consider a quasi - projective variety $Y \subseteq \mathbb{P}^n$.

Definition 1.3.3. A function $f : Y \to k$ is regular at a point $P \in Y$ if there is an open neighborhood U with $P \in U \subseteq Y$, and homogeneous polynomials $g, h \in S = k[x_0, \ldots, x_n]$, of the same degree, such that h is nowhere zero on U, and $f = g/h$ on U. (Note that in this case, even though g and h are not functions on $\mathbb{P}^n$, their quotient is a well - defined function whenever $h \neq 0$, since they are homogeneous of the same degree.) We say that f is regular on Y if it is regular at every point.

Remark 1.3.4. As in the quasi - affine case, a regular function is continuous (proof left to reader). An important consequence of this is the fact that if f and g are regular functions on a variety X, and if $f = g$ on some nonempty open subset $U \subseteq X$, then $f = g$ everywhere. Indeed, the set of points where $f - g = 0$ is closed and dense, hence equal to X.

Now we can define the category of varieties.

Definition 1.3.5. Let k be a fixed algebraically closed field. A variety over k (or simply variety) is any affine, quasi - affine, projective, or quasi - projective variety as defined above. If X, Y are two varieties, a morphism $\varphi : X \to Y$

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